

Ratios & Simplest Form

A ratio compares two quantities. Learn how to write one, reduce it to simplest form, and decide when two figures (or recipes, or mixtures) are proportional.

What is a ratio?

The idea: A **ratio** $a : b$ compares two quantities — for every a units of the first there are b units of the second. The numbers a and b are called the **terms** of the ratio.

Width : height of an image = 60 : 40

Simplest form

Divide both terms by their highest common factor (HCF). The ratio's meaning doesn't change.

$$72 : 96 = (72 \div 24) : (96 \div 24) = 3 : 4$$

Proportional (similar) quantities

Two ratios are **proportional** when they are equal in their simplest forms. Rectangles are **similar** when their width : height ratios match.

- 60 : 40 and 30 : 20 are both 3 : 2 in simplest form, so those rectangles are similar.
- A square 60 : 60 is 1 : 1 — not 3 : 2 — so it looks different (distorted).

Watch the units: compare like with like. 1 kg : 800 g is really 1000 g : 800 g = 5 : 4, not 1 : 800.

Prerequisite refresher: "HCF" is the largest number that divides both terms. Reducing a ratio is like reducing a fraction.

Where marks slip away

Trap 1 — Mixed units. Convert both terms to the same unit before writing a ratio (1 m : 50 cm = 100 cm : 50 cm = 2 : 1).

Trap 2 — Order matters. 3 : 2 is not the same as 2 : 3. Write the terms in the order asked.

Trap 3 — Reduce fully. 24 : 18 is not simplest; divide by 6 to get 4 : 3.

Trap 4 — Adding to terms breaks proportion. Adding the same number to both terms of a ratio usually changes it ($3 : 30$ is $1 : 10$, but $12 : 39$ is $4 : 13$).

Slip 5 — A ratio has no units. After simplifying, write just the numbers ($4 : 3$), not " $4 \text{ cm} : 3 \text{ cm}$ ".

Model answers — write them like this

Example A. Write $600 : 900$ in its simplest form.

Step 1: HCF of 600 and 900 is 300.

Step 2: $600 \div 300 = 2$; $900 \div 300 = 3$.

$\therefore 2 : 3$

Example B. Are the ratios $3 : 4$ and $72 : 96$ proportional?

Step 1: $3 : 4$ is already simplest.

Step 2: $72 : 96 = 3 : 4$ (divide both by 24).

Step 3: Both simplest forms are $3 : 4$, so they are proportional.

$\therefore$ Yes, proportional

Example C. A rectangle is 60 mm wide and 40 mm high. Is it similar to a $30 \text{ mm} \times 20 \text{ mm}$ rectangle?

Step 1: $60 : 40 = 3 : 2$; $30 : 20 = 3 : 2$.

Step 2: The width : height ratios are equal.

$\therefore$ Yes, the rectangles are similar

Example D. Anagh mixes 600 mL orange juice with 900 mL apple juice. Write the ratio in simplest form.

Step 1: Orange : apple = $600 : 900$.

Step 2: Divide both by 300: $2 : 3$.

$\therefore 2 : 3$